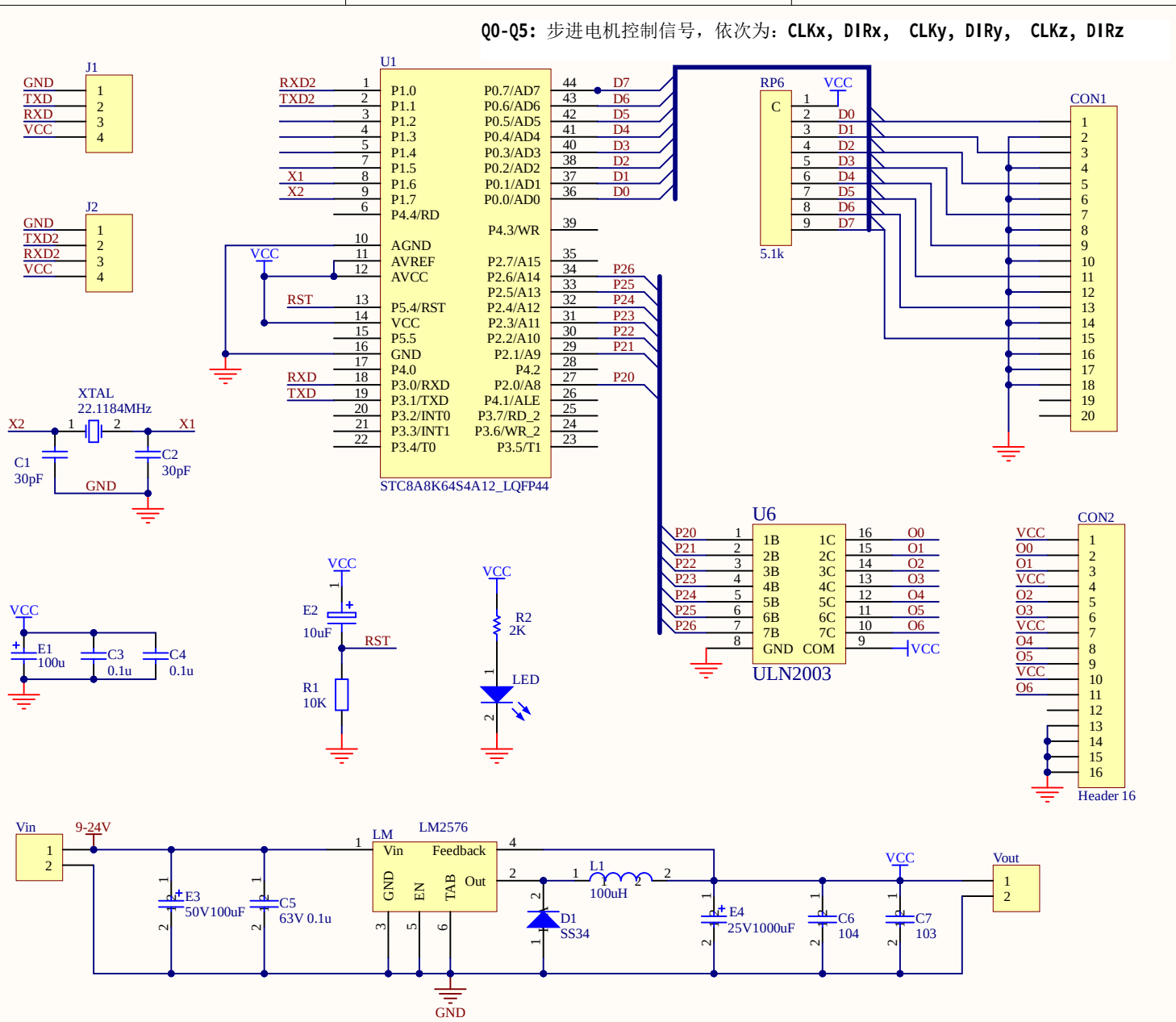

Q0-Q5: 步进电机控制信号，依次为：CLKx, DIRx, CLKy, DIRy, CLKz, DIRz
J1
GND
TXD
RXD
VCC
J2
GND
TXD2
RXD2
VCC
U1
RXD2
TXD2
X1
X2
RST
RXD
TXD
P1.0
P1.1
P1.2
P1.3
P1.4
P1.5
P1.6
P1.7
P4.4/RD
AGND
AVREF
AVCC
P5.4/RST
VCC
P5.5
GND
P4.0
P3.0/RXD
P3.1/TXD
P3.2/INT0
P3.3/INT1
P3.4/T0
P0.7/AD7
P0.6/AD6
P0.5/AD5
P0.4/AD4
P0.3/AD3
P0.2/AD2
P0.1/AD1
P0.0/AD0
P4.3/WR
P2.7/A15
P2.6/A14
P2.5/A13
P2.4/A12
P2.3/A11
P2.2/A10
P2.1/A9
P4.2
P2.0/A8
P4.1/ALE
P3.7/RD_2
P3.6/WR_2
P3.5/T1
STC8A8K64S4A12_LQFP44
D7
D6
D5
D4
D3
D2
D1
D0
P26
P25
P24
P23
P22
P21
P20
RP6
C
5.1k
VCC
D0
D1
D2
D3
D4
D5
D6
D7
CON1
XTAL
22.1184MHz
X2
X1
C1
30pF
C2
30pF
GND
U6
P20
P21
P22
P23
P24
P25
P26
1B
2B
3B
4B
5B
6B
7B
GND
1C
2C
3C
4C
5C
6C
7C
COM
O0
O1
O2
O3
O4
O5
O6
VCC
ULN2003
CON2
VCC
E1
100u
C3
0.1u
C4
0.1u
E2
10uF
RST
R1
10K
R2
2K
LED
Vin
9-24V
LM
LM2576
Vin
Feedback
GND
EN
TAB
Out
E3
50V100uF
C5
63V 0.1u
L1
100uH
D1
SS34
E4
25V1000uF
C6
104
C7
103
Vout
Header 16
Title *
Size: A4
Number:*
Revision:*
Date: 2025/5/26
Time: 15:37:49
Sheet * of *
File: G:\绿联家庭云盘\1教学\1综合设计项目（项目1）\0升级版\PCB(Upgrade)\MotorControl.SchDoc
Altium

Comment	Description	Designator	Footprint	LibRef	Quantity
	Capacitor	C1, C2	RAD0.1	cap	2
0.1u	Capacitor (Semiconductor SIM Model)	C3, C4	RAD-0.2	Cap Semi	2
63V 0.1u	Capacitor	C5	RAD0.2	CAP	1
104	Capacitor	C6	RAD0.2	CAP	1
103	Capacitor	C7	RAD0.2	CAP	1
	Header, 20-Pin	CON1	HDR2X10	HEADER 20	1
Header 16	Header, 16-Pin	CON2	HDR2X8	Header 16	1
SS34	Default Diode	D1	SS34	SS34	1
100u	Polarized Capacitor (Surface Mount)	E1	RAD-0.2	Cap Pol3	1
10uF		E2	RAD-0.2	Pro77.tmp_47u_27	1
Cap Pol3	Polarized Capacitor (Surface Mount)	E3, E4	RAD0.2C	Cap Pol3	2
Header 4	Header, 4-Pin	J1, J2	HDR1X4	Header 4	2
Inductor	Inductor	L1	RAD0.3C	Inductor	1
	Typical INFRARED GaAs LED	LED	BAT-2	LED0	1
LM2576		LM	TO-263-5	LM2576	1
Vin	Header, 2-Pin	Power0	POWER2	Header 2	1
Vout	Header, 2-Pin	Power1	POWER2	Header 2	1
22k	Resistor	R1	AXIAL-0.3	res2	1
	Resistor	R2	AXIAL-0.3	RES1	1
5.1k	RESPACK8	RP6	HDR1X9	respack8	1
STC8A8K64S4A12_LQFP44		U1	LQFP44_STC8A8K	STC8A8K64S4A12_LQFP44	1
ULN2003		U6	DIP-16	Pro286.tmp_MC1416_21	1
22.1184MHz	Crystal Oscillator	XTAL	RAD-0.2	XTAL	1